NIELSEN / SOLUTION ENGINEERING

Conversational **Business Intelligence**

A practical conversational layer over enterprise semantic models - designed to turn natural-language questions into grounded analytics.



THE WORK

React experience • Node.js BFF • MCP integration • DAX-backed Power BI analysis

PROJECT SHOWCASE / OCTOBER 2026

01 / OPPORTUNITY

From questions to decisions

A single analytics experience designed around how business users actually ask for information.

THE CHALLENGE

Power BI semantic models contain trusted metrics, but a user still needs to know where to look, which model to choose, and how to translate a business question into a useful report.

OUR RESPONSE

A chat workspace keeps the semantic model explicit and uses the existing MCP-backed analytics service.

EXPERIENCE

Ask naturally

Guided model selection, conversational answers, charts, tables, and export in one workspace.

INTEGRATION

Reuse the pipeline

A BFF validates and adapts requests while the MCP Host and Server perform analytics.

CONTROL

Make it governable

Clear boundaries for model access, identity, operational telemetry, and future hardening.

WHAT THIS SHOWCASE COVERS

1. A polished user journey from question to evidence-backed answer
2. The application architecture and the client-side integration points
3. Current implementation plus a credible path to enterprise operations

THE DESIGN IN ONE LINE

A focused BI experience that connects questions to governed analytical evidence.

02 / EXPERIENCE

One workspace, one question

The interface reduces the distance between a business prompt and a usable analytical result.

Conversational BI Workspace

SEMANTIC MODEL

Schedule Completeness TSG

What was the completeness trend in Q1? >

YOU ASKED

Which customers had the biggest improvement, and what changed month to month?

ANALYSIS

The response combines a concise explanation with a structured result that can be reviewed or exported.

Customer A	76%	91%	**+15 pts**
Customer B	83%	94%	**+11 pts**
Customer C	88%	96%	**+8 pts**

Illustrative content - not production data

FOUR MOMENTS THAT MATTER

1. **Choose** — Select trusted data
2. **Ask** — Use business language
3. **Review** — Read chart + table
4. **Act** — Export and share

03 / ARCHITECTURE

A composable analytics stack

The UI and BFF add a business-friendly experience without replacing the existing analytical execution path.



04 / WORKFLOW

How an answer is produced

A request crosses defined layers, then returns as a readable result with analytical evidence.



WHERE INTELLIGENCE IS APPLIED

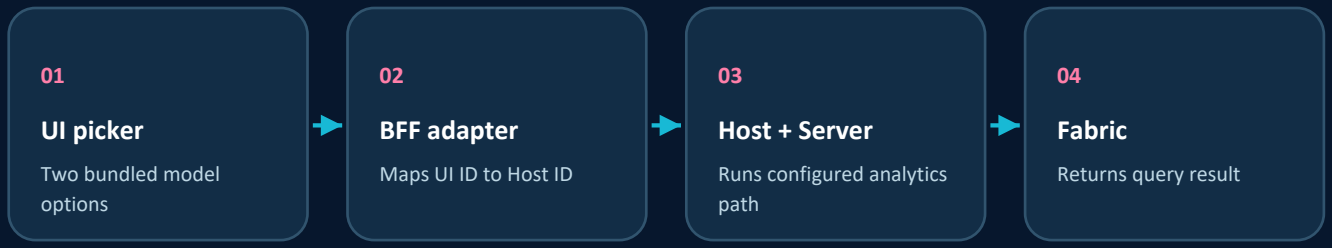
The Host coordinates the conversation and MCP tools. In the observed server code, an LLM-assisted step creates DAX before the Power BI query tool runs. This is not a direct browser-to-Fabric call.

05 / MODEL GOVERNANCE

The semantic model is deliberate

The selected model is a business context decision, not an accidental side effect of the prompt.

CURRENT MODEL JOURNEY



Today the UI defines Schedule Completeness TSG and Mapping Stats VOD. New Fabric models do not automatically appear in the picker; the catalogue and deployed execution mapping need coordinated updates.

WHY THIS MATTERS

CLARITY

The answer has a scope

A chosen model helps users understand which measures and dimensions underpin the result.

GOVERNANCE

Mappings can be controlled

The BFF can validate an approved catalogue and translate display IDs to the Host contract.

EVOLUTION OPPORTUNITY

Discover approved models through the MCP Host, cache a governed catalogue, and populate the UI from one source of truth. This is a planned improvement, not current automatic discovery.

06 / OUTPUTS

Answers built for review

The result is more than generated prose: users can inspect the analytical shape and move the data onward.

ANALYSIS / COMPLETENESS BY MONTH

MODEL: TSG

SUMMARY

Completeness rose across the period. The largest movement came from the first month to the second; review the underlying rows before using the result in a decision.

Illustrative answer layout

TREND VISUAL



STRUCTURED RESULT

Month	Customer	Completeness	Change
Jan	Customer A	76%	—
Feb	Customer A	84%	+8 pts
Mar	Customer A	91%	+7 pts

SHAREABLE FORMS

Narrative

Plain-language explanation

Visual

Trend and comparison views

Table

Rows that can be inspected

CSV

Portable data export

07 / TRUST

Security is a boundary, not a label

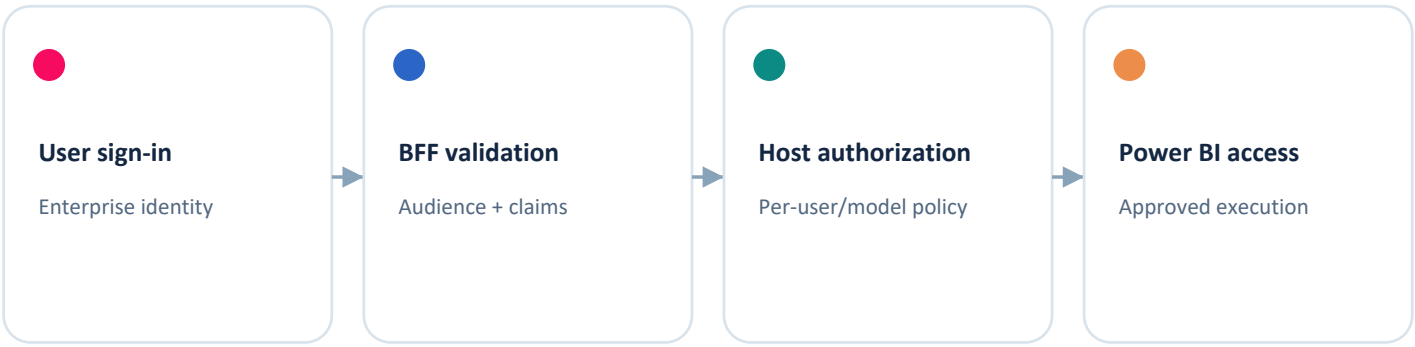
The current no-Entra edition keeps identity limitations explicit while providing a route to a per-user enterprise design.

CURRENT ACCESS BOUNDARY

The BFF uses one shared workspace identity for reachable users. Browser UUIDs separate anonymous sessions and rate-limit buckets; they are not proof of identity or model authorization.

1. **Network edge**
   Protect the no-login app behind approved SSO gateway, VPN, or private network.
2. **Application edge**
   Validate request shape, apply rate limits, and keep secrets server-side.
3. **Analytics edge**
   Use configured service credentials to query the semantic model.

PRODUCTION IDENTITY PATH



DESIGN PRINCIPLE

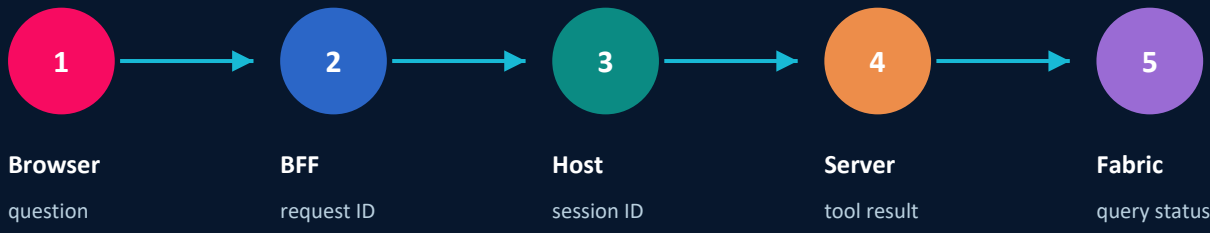
A session ID is conversation state. An authenticated identity is an authorization input. Production history and semantic-model permissions should be keyed to the latter, not retroactively inferred from the former.

08 / OPERATIONS

Designed to be observable

The application already has useful diagnostic seams; production operations turn them into a measurable service.

REQUEST TRACE



Carry a correlation ID across boundaries so an operator can trace a slow or failed answer without exposing query contents in routine logs.

SERVICE HEALTH CHECKLIST

Availability	Health/readiness checks; ingress and upstream reachability
Latency	End-to-end time and each call to Host, Server, and Fabric
Failures	Timeouts, invalid model IDs, expired sessions, query errors
Quality	Answer rendering, DAX normalization, table/chart integrity
Security	Auth events, rejected input, secret rotation, access reviews

Operational measures above are the target service scorecard, not claimed production results.

09 / RESILIENCE

A clear path to scale

A horizontally scalable application depends on externalizing state before adding replicas.

TODAY: FAST TO ITERATE

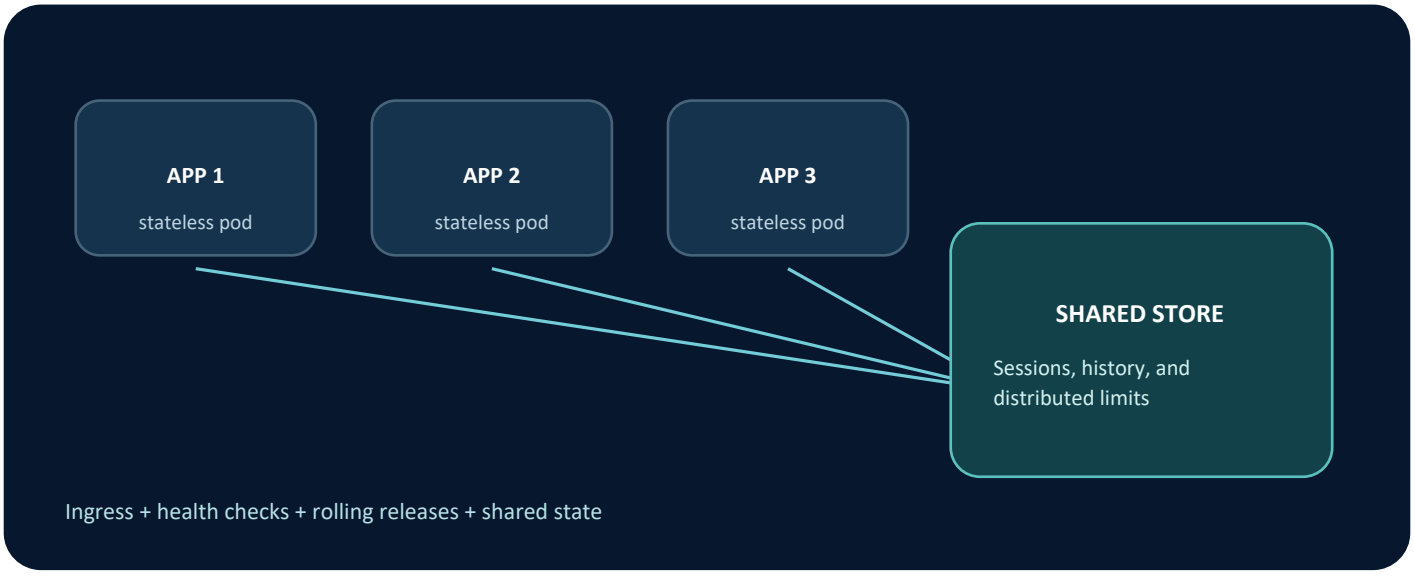
The BFF and MCP Host retain some session and rate-limit state in process memory. This is practical for the current edition, but a restart or multiple replicas can separate a follow-up from its context.

Single app instance

In-memory state

Restart clears state

PRODUCTION DESIGN



Then replicas can be scaled with demand and upgraded incrementally. Kubernetes Deployments and autoscaling are relevant patterns, but need client infrastructure and load testing before an availability claim.

10 / LIFECYCLE

Upgrade without losing confidence

A maintainable integration isolates contracts, configuration, and recoverable state.

01

CONTRACT

Version the BFF-to-Host request and response schema; keep adapter tests for real payload shapes.

02

CONFIGURATION

Keep model mappings, endpoints, credentials, and feature flags out of the browser and source code.

03

RELEASE

Promote tested images, run smoke tests, monitor error rates, and retain a rollback route.

04

RECOVERY

Back up durable session/history stores and settings; verify restore, retention, and ownership.

PRACTICAL RECOVERY TARGET

Back up what must survive a restart; rehearse restore; document who approves a model or endpoint change. The current process-local stores are not a durable backup solution.

11 / REUSE

A foundation for the next engagement

The reusable pattern is the separation of experience, orchestration, and governed analytics.

01

Experience layer

Adapt the chat, model selection, visual response, and guidance to a new business domain.

02

Integration layer

Reuse the BFF contract and response adaptation against approved MCP-backed services.

03

Governance layer

Introduce client identity, model policy, telemetry, and operational runbooks at defined seams.

TAKEAWAY

An understandable BI interface can sit over an established analytics stack - with integration boundaries that support future security and scale.

Selected technical references

Model Context Protocol Power BI semantic models AWS Bedrock Claude